Higher Diploma in Computer Science

Computer Systems & Networks

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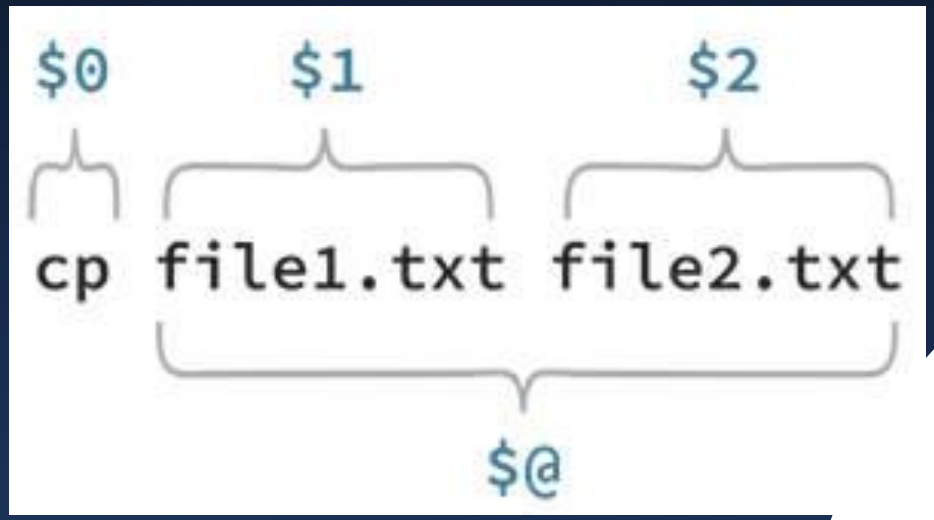
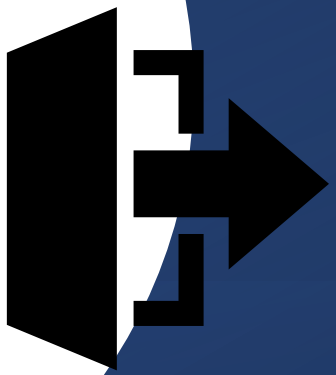


Positional Parameters

Exit codes



Positional parameters are **special variables** in shell scripting that store **arguments passed to a script or function**, based on their **position**

An **exit code** is a **number returned by a process or command** when it finishes running (tells system whether the *command succeeded or failed*)

```
#!/bin/bash
variable=value
```

Variables: RECAP

Variables

- Variables are a key part of any programming language.
- A very simple use of variables is shown here:

```
echo Enter your first number:
read first
echo Enter your second number:
read second
z=`expr $first + $second`
echo The sum is ${z}
```

Variables

- Another way to assign to a variable is to use the output of another command as the contents of the variable by enclosing the command in backtick ` characters:

```
#!/bin/bash
CURRENT_DIRECTORY=`pwd`
echo "You are in $CURRENT_DIRECTORY"
```

- It is also possible to get input from the user of the script and assign that input to a variable using the `read` command:

```
echo -n "What is your name?"
read name
echo "Hello ${name}! Nice of you to join us
```

What syntax is missing that causes an error when executed?

Variables: Summary

- **Don’t need to declare**

```
caroline@caroline-VirtualBox:~$ x=10
caroline@caroline-VirtualBox:~$ echo $x
```

- **read command**

```
caroline@caroline-VirtualBox:~$ read number
45
caroline@caroline-VirtualBox:~$ echo $number
```

- **No type**

```
caroline@caroline-VirtualBox:~$ x=8
caroline@caroline-VirtualBox:~$ y=5
caroline@caroline-VirtualBox:~$ z=$x+$y
caroline@caroline-VirtualBox:~$ echo $z
```

The output is:

- **expr**

```
x=8
y=5
z=$x+$y
echo $z
```

```
expr $x+$y
```

```
z=`expr $x + $y`
echo The sum is $z
```

Now try out multiplying 6 by 4

Command Chaining

Command Chaining ; && ||

; (semicolon) → run one command right after another

```
$ls ; pwd ; whoami
```

&& (and) → second command to only run if the first command is successful

```
$mkdir compsys2025 && cd compsys2025
```

*With **;** the second command will run even if the first one fails*

Command Chaining ; && ||

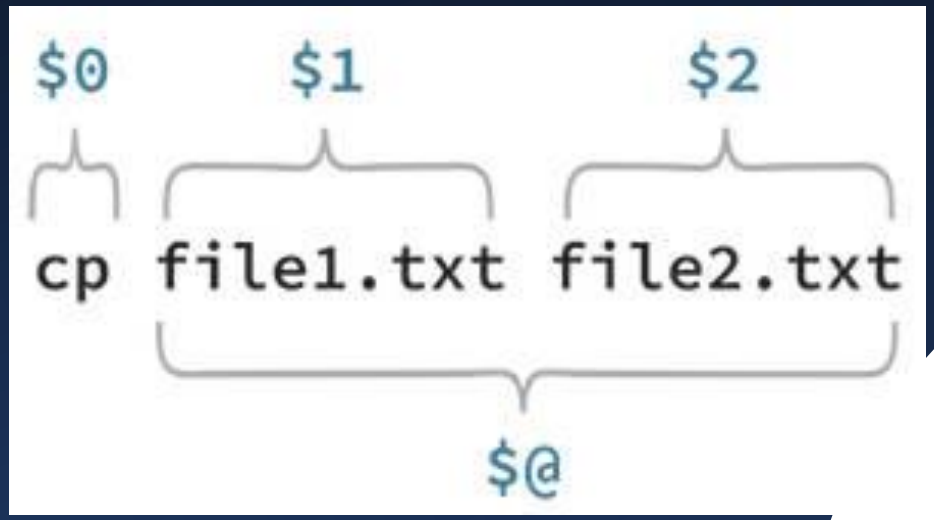
Check if a directory exists **[-d "/path/dir/"]**

```
compsys@compsys-virtualbox:~$ [ -d "temp" ] && echo "Directory exists"
Directory exists
compsys@compsys-virtualbox:~$ [ ! -d "rainy" ] && echo "Directory DOESN'T exists"
Directory DOESN'T exists
```

|| (or) →Here, you want to execute second command only if first command ***is not*** successful

```
$ [ -d "newHDipFolder" ] && echo "Directory exists" || mkdir ~/newHDipFolder
```

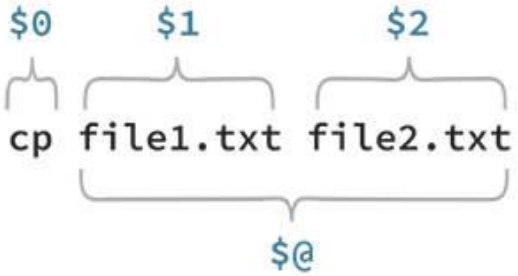
`-b filename` - Block special file
`-c filename` - Special character file
`-d directoryname` - Check for directory Existence
`-e filename` - Check for file existence, regardless of type (node, directory, socket, etc.)
`-f filename` - Check for regular file existence not a directory
`-G filename` - Check if file exists and is owned by effective group ID
`-G filename set-group-id` - True if file exists and is set-group-id
`-k filename` - Sticky bit
`-L filename` - Symbolic link
`-O filename` - True if file exists and is owned by the effective user id
`-r filename` - Check if file is a readable
`-S filename` - Check if file is socket
`-s filename` - Check if file is nonzero size
`-u filename` - Check if file set-user-id bit is set
`-w filename` - Check if file is writable
`-x filename` - Check if file is executable



Positional Parameters

Positional parameters are **special variables** in shell scripting that store **arguments passed to a script or function**, based on their **position**

Variables → Positional Parameters



- The **$0** variable contains the name of the script itself. Additionally, you can pass *arguments* to your script:

```
compsys@compsys-virtualbox:~$ cat pospa*
#!/bin/bash
echo -n "What's your name? "
read name
echo "Hello ${name}! You're looking at your file $0"
```

A dollar sign followed by a number *N* corresponds to the **Nth** argument passed to the script

Positional Parameters: $

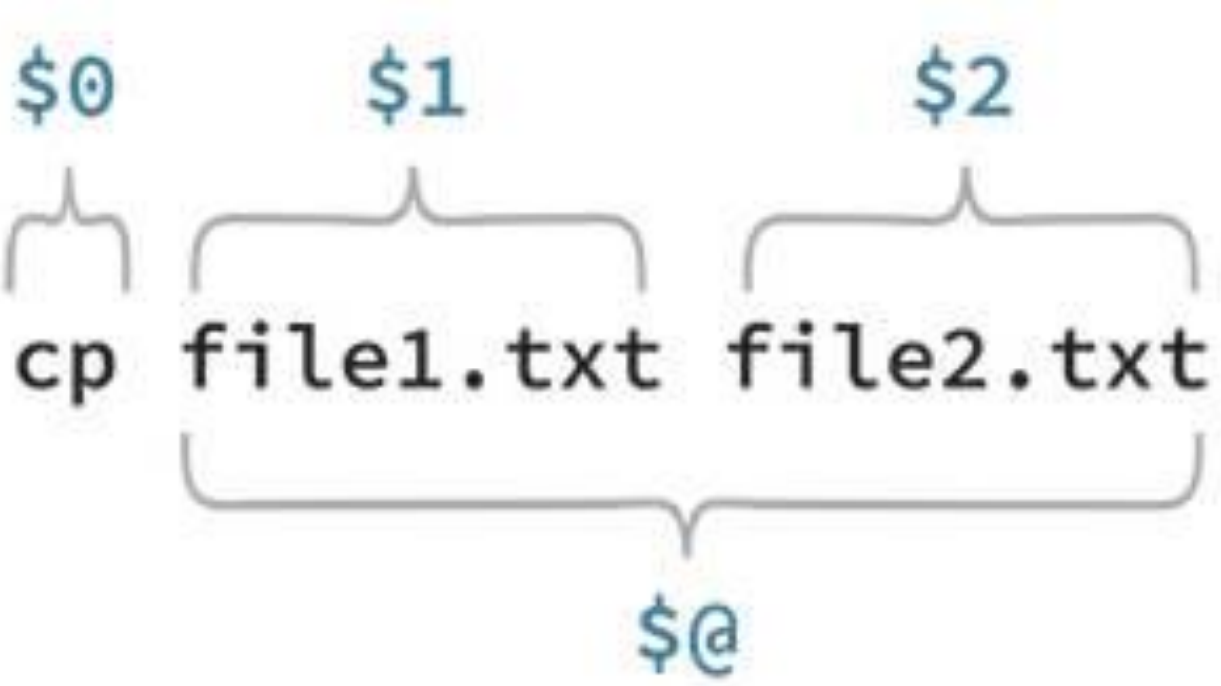
…referenced using the **$** sign.

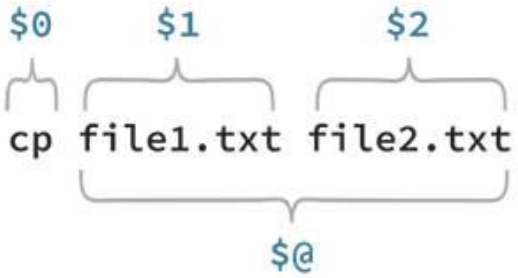
$1 variable is used to read the **first** command line argument

$2 variable is used to read the **second** command line argument etc.

(Two digits are written as **${10}, ${11}** and so on.)

$0 is the **script name** or else the **command** given

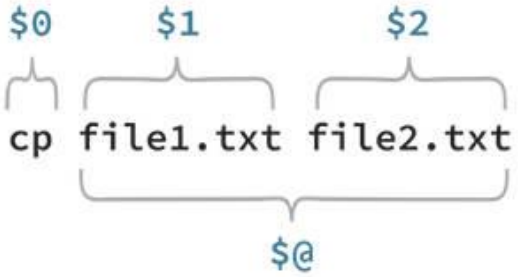




Positional Parameters: $#

$# the number of positional parameters, excluding **$0**.

$# is often used to check that the user has entered the correct number of command line arguments were passed to the script.

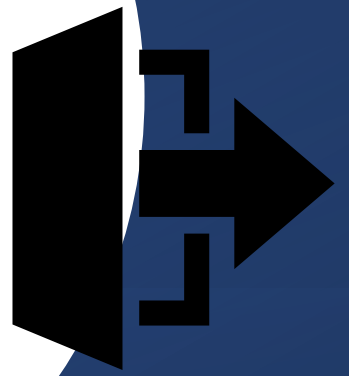


Positional Parameters: $*

$* references ***all*** the arguments passed to a program

(useful in programs that take an indefinite or variable number of arguments)

Exit codes



An **exit code** is a **number returned by a process or command** when it finishes running (tells system whether the *command succeeded or failed*)

Exit Codes

indicates the response from the command or a script after execution

- help us determine whether a process ran Successfully

For the bash shell's purposes, a command which exits with a:

- zero (0) exit status = **success;**
- non-zero (1-255) exit status =**failure**

command causes the shell or program to terminate

- You can set the exit code of your own script with the **exit** command.

$?

When running commands at a shell prompt, the special variable **$?** contains a number that indicates the **result of the last command** executed

0 → ✅ **Success**

Non-zero → ❌ **Failure or error (**A value of 1 generically indicates that some error has happened)

You can check the exit code of the **last executed command** using: **$echo $?**

```
compsys@compsys-virtualBox:~$ cat employee.txt
compsys@compsys-virtualBox:~$ echo $?
0
compsys@compsys-virtualBox:~$ grep -q asdfjkl; /etc/passwd
compsys@compsys-virtualBox:~$ echo $?
1
```

Use **$?** variable to see if the previous command completed successfully (no errors=0)

```
if [ ! -f compsys.txt ];
then
  echo The file does not exist and I display an error message
  exit 23      #random code for error condition
fi

echo The file exists and we will do something else
echo “here you could have your script doing something"

exit 0
```

$./myexit_test.sh
The file does not exist and I display an error message

$echo $?
64

$touch compsys.txt

$./ myexit_test.sh
The file exists and we will do something else
here you could have your script doing something

$echo $?
0

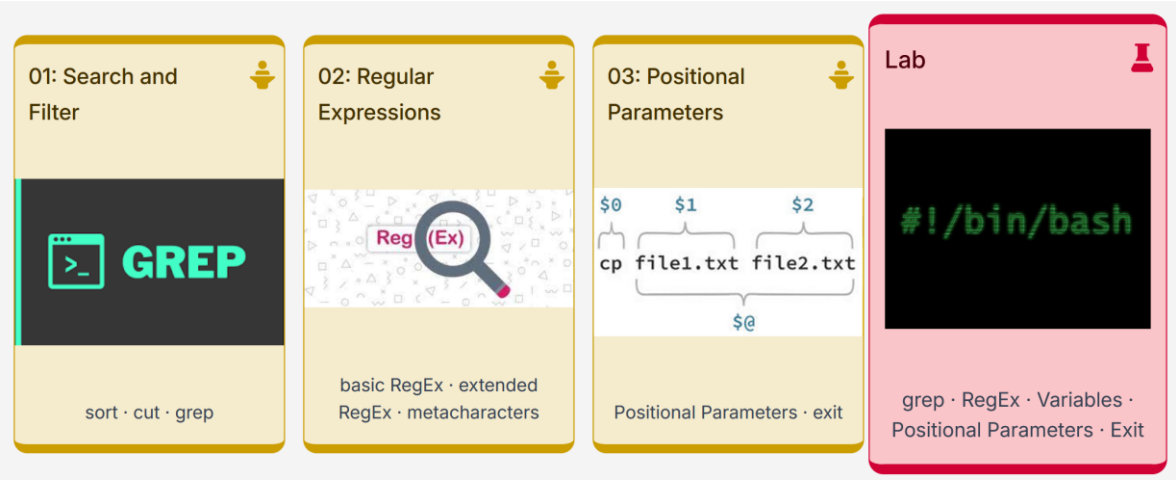
PRACTICE: Using positional parameters & redirection to append to fields in file

- Write a script that can be used to add in a new employees details to the **employee.txt** file

```
Caroline@ICTSkills>~$ cat employee.txt
100 frank manager sales €5000
200 caroline developer technology €5500
300 mary sysadmin technology €7000
400 rosanne manager marketing €9500
500 eamonn dba technology €6000
```

- What could be the **steps** you might take with this?

Try out this weeks Lab



Manipulate Files & Contents

01. sort, cut

02. grep

03. Variables

04. Positional Parameters

Cisco NetAcad

- $* references all the arguments passed to a program

```
#!/bin/bash
echo "There were $# variables passed "
echo "All arguments passed can be viewed using the \$* variable E.G. $* "
```

EXERCISE: Practice your coding...

Q: what is **let**? Try **man let**; do you need to use **help let**?

```
#!/bin/bash


echo -e  "\$1=$1"
echo -e "\$2=$2"


let add=$1+$2
let sub=$1-$2
```